

LOS ANGELES | MAY 22, 2024

aws SUMMIT



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AIM201

Accelerate generative AI application development with Amazon Bedrock

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Agenda

- Introduction
- Choice of models
- Customization to your needs
- Integration with your workflows
- Security and governance
- Consumption options
- Resources

Building generative AI applications is challenging



Accessing
multiple FMs
and newer
versions



Customizing
FMs is not easy



Data privacy
and security



Getting FMs
to execute tasks



Connecting to
data sources or
corporate
documents



Difficult
to manage
infrastructure



Amazon Bedrock

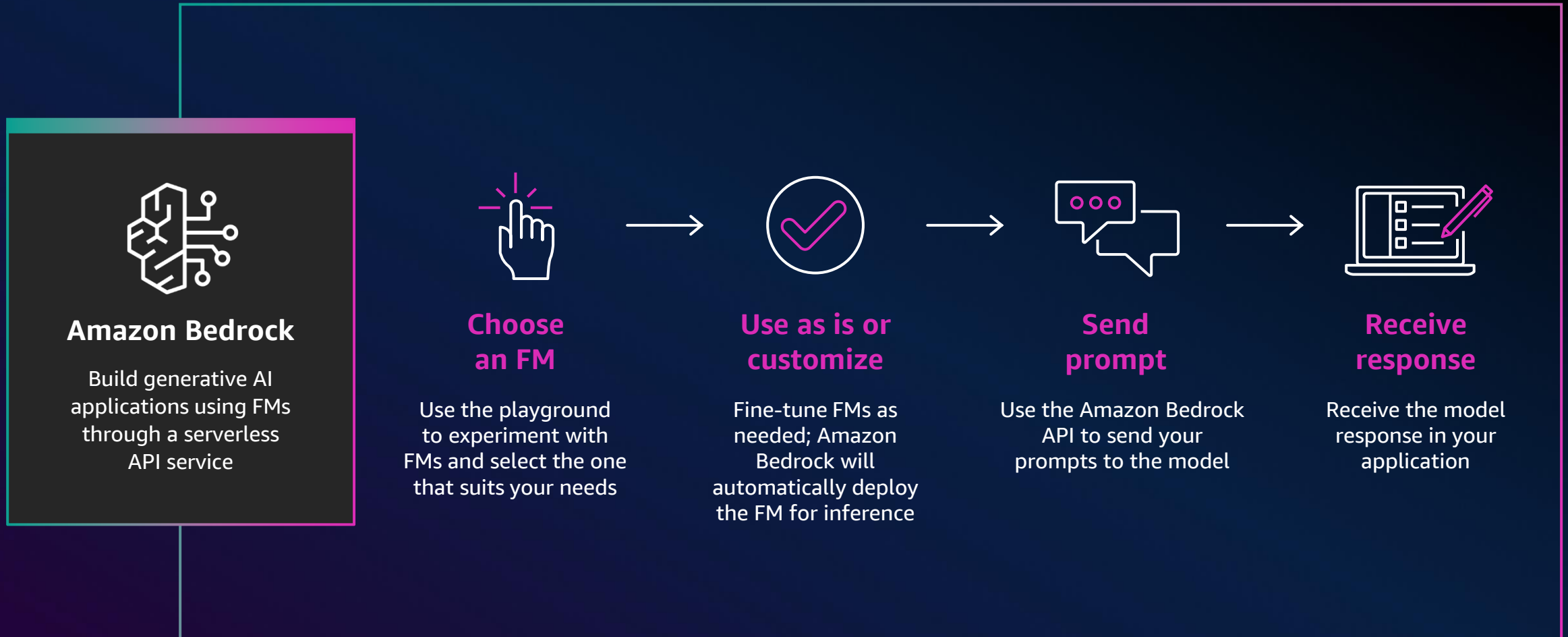
The easiest way to build and scale generative AI applications with foundation models

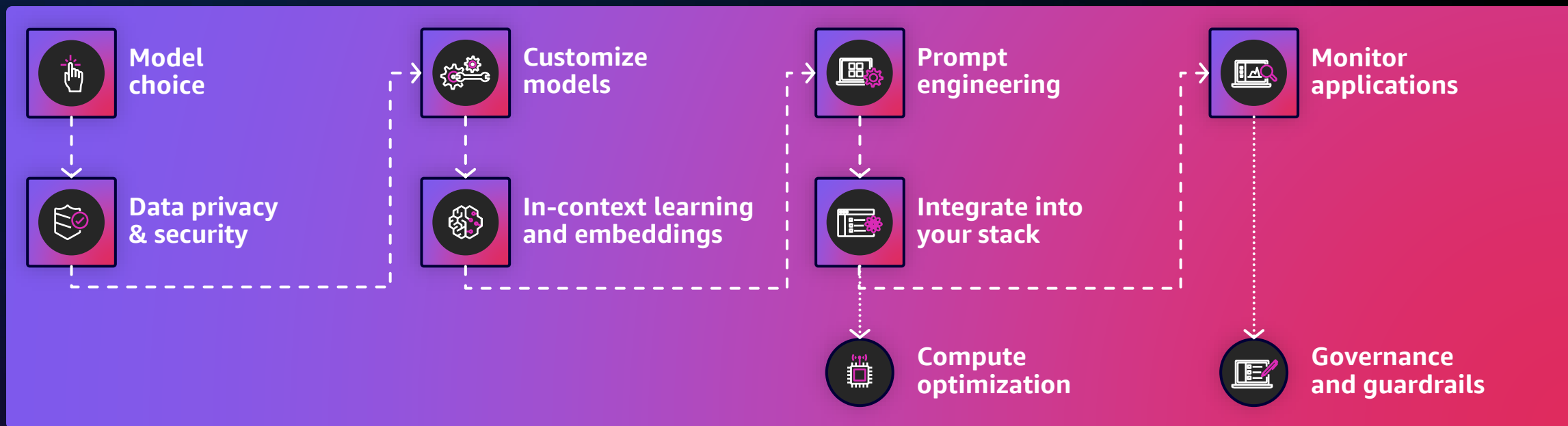
Choice of industry-leading FMs available via a single API

Customize your models using your organization's data

Enterprise-grade security and privacy

How Amazon Bedrock works





Amazon Bedrock provides capabilities to accelerate generative AI application development

SECURE

Capabilities to serve your data protection and privacy needs; support for compliance

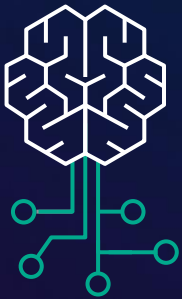
SCALABLE

Fully managed service that scales with demand so you don't have to manage infrastructure

SAFE

Comprehensive tooling and resources to build AI responsibly across the application lifecycle

Amazon Bedrock simplifies



Choice



Customization



Integration



Responsible AI

Amazon Bedrock

Broad choice of models

AI21labs

amazon

ANTHROPIC

cohere

Meta

MISTRAL AI

stability.ai

Contextual
answers,
summarization,
paraphrasing

Text summarization,
generation, Q&A, search,
image generation

Summarization,
complex reasoning,
writing, coding

Text generation,
search, classification

Q&A and reading
comprehension

Text summarization,
Q&A, text classification,
text completion, code
generation

High-quality
images and art

Jurassic-2 Ultra

Amazon Titan Text Lite

Claude 3 Opus

Command

Llama 3 8B

Mistral Large

Stable Diffusion XL1.0

Jurassic-2 Mid

Amazon Titan Text Express

Claude 3 Sonnet

Command Light

Llama 3 70B

Mistral 7B

Stable Diffusion XL 0.8

Amazon Titan Text Premier

Claude 3 Haiku

Embed English

Llama 2 13B

Mixtral 8x7B

**Amazon Titan Text
Embeddings**

Claude 2.1

Embed Multilingual

Llama 2 70B

**Amazon Titan Text
Embeddings V2**

Claude 2

Command R+

Claude Instant

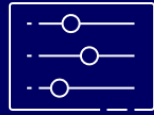
Command R

**Amazon Titan Multimodal
Embeddings**

**Amazon Titan Image
Generator**

Amazon Titan Text Embeddings V2

Flexible embeddings model optimized for RAG



Flexible dimension sizes



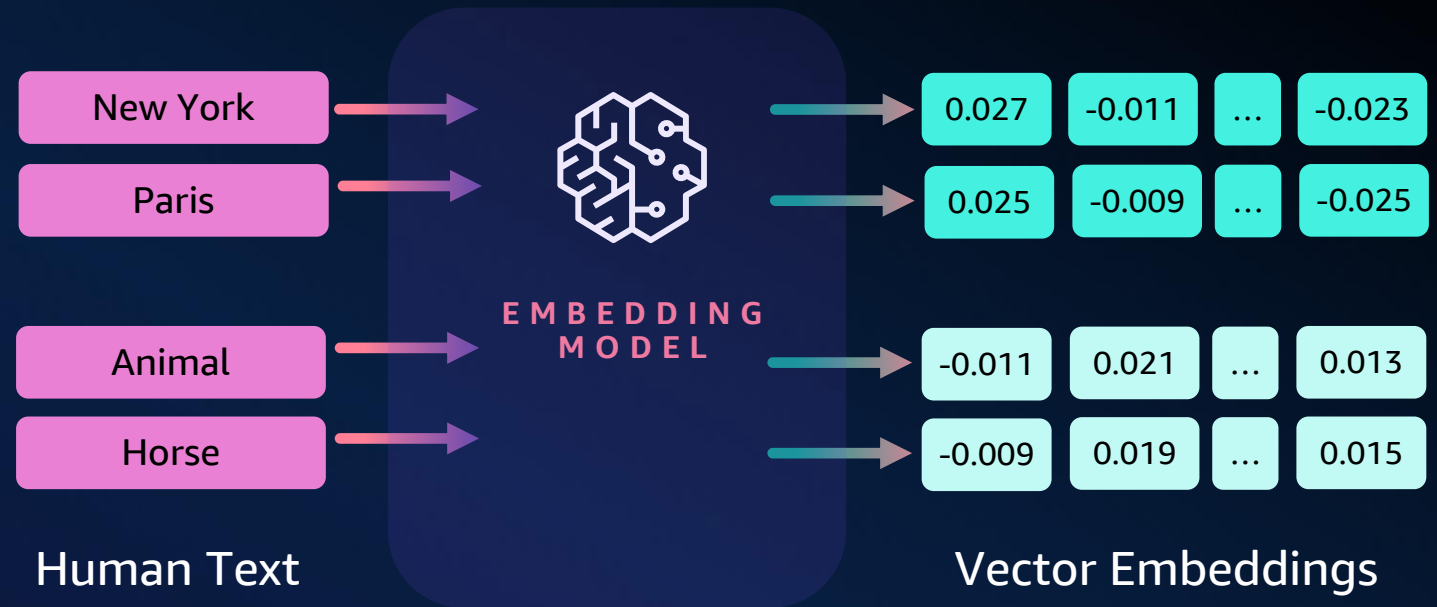
Optimized for RAG



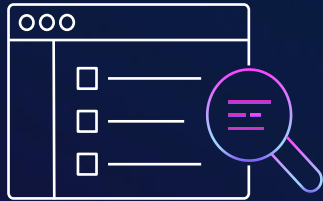
100+ pretrained languages

What are embeddings?

- Numerical representation of text (vectors) that captures semantics and relationships between words.
- Embedding models capture features and nuances of the text.
- Rich embeddings can be used to compare text similarity.
- Multilingual Text Embeddings can identify meaning in different languages.



Why are embeddings important for RAG?



Powers **text retrieval based on semantic meaning**



Used to augment prompts with **more accurate context** from vector stores using Retrieval Augmented Generation (RAG)



High-accuracy embeddings leads to improved context **and higher quality LLM-generated** responses to a user query

Amazon Titan Text Embeddings V2



Amazon Titan Text Embeddings V2

Translates text inputs into numerical representations (embeddings)

Input tokens: **512** (expandable to 8K)

Output vectors: **Flexible** (1,024, . . .)

Language: **Multilingual** (100 languages)



Highlights

- Amazon Titan Text Embeddings V2 offers fast, cost-effective, high-performance, accurate embeddings in 100 languages
- Optimized for RAG, text-retrieval tasks, semantic similarity, classification, and clustering
- Flexible embeddings (1,024, 512, 256) with 97% accuracy retention for retrieval from 1,024 to 512

Model evaluation



Playground

Evaluate directly in the playground as you try out different models; compare across cost latency and usage dimensions

Use playground as you narrow down on the use case and identify the FM



Programmatic

Evaluate as part of your application development lifecycle or model customization; metrics include accuracy, toxicity, and robustness

Use programmatic evaluation as you iterate on the use case or the model



Human-in-the-loop Bring your own team

Use the framework offered to organize your team for your evaluation; better suited for subjective criteria such as brand voice, clarity, and tone

Bring your own team as you start testing your first prototype or get ready for pilot



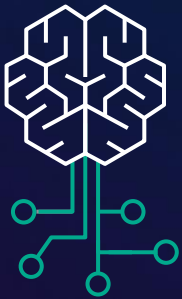
Human-in-the-loop AWS Managed team

Use AWS expertise for an expert evaluation, in case you don't have the budget or expertise in-house

AWS-managed team as you get ready for production launch of your application



Amazon Bedrock simplifies



Choice



Customization



Integration

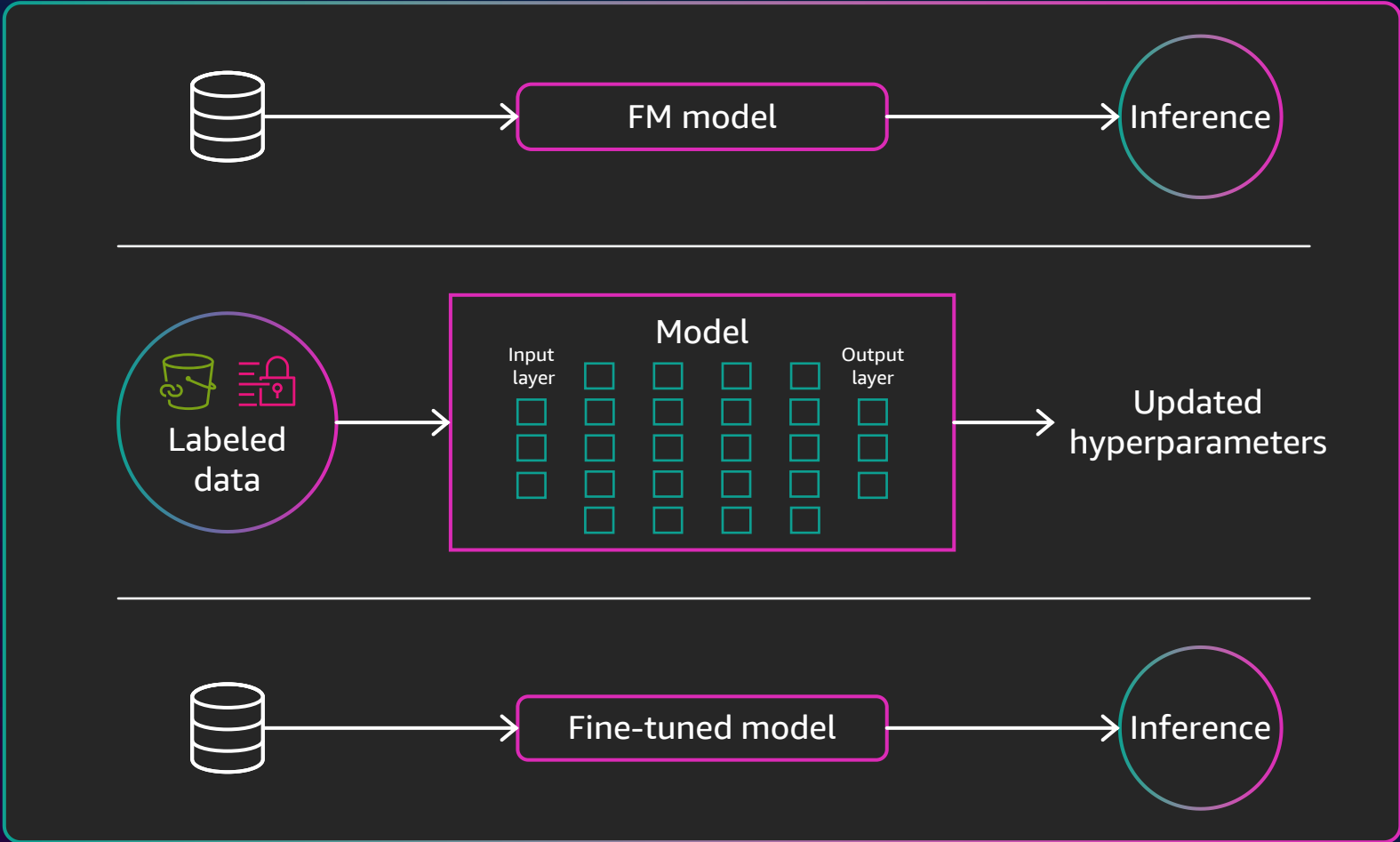


Responsible AI

Fine-tuning



Customize
models



Customize models with simple configuration

Epochs, batch size, learning
rate, warmup steps

Hyperparameters [Info](#)

Epochs
The total number of iterations of all the training data in one cycle for training the model.

Batch size
The number of samples processed before the model is updated.

Learning rate
The step size for incrementing parameters at each iteration.

Learning rate warmup steps
Number of iterations over which learning rate is gradually increased to the initial rate specified.

- ✓ Easy
- ✓ Secure
- ✓ Private

Simplified
hyperparameters update

Data encrypted
with CMK

Access to S3 bucket
via AWS PrivateLink

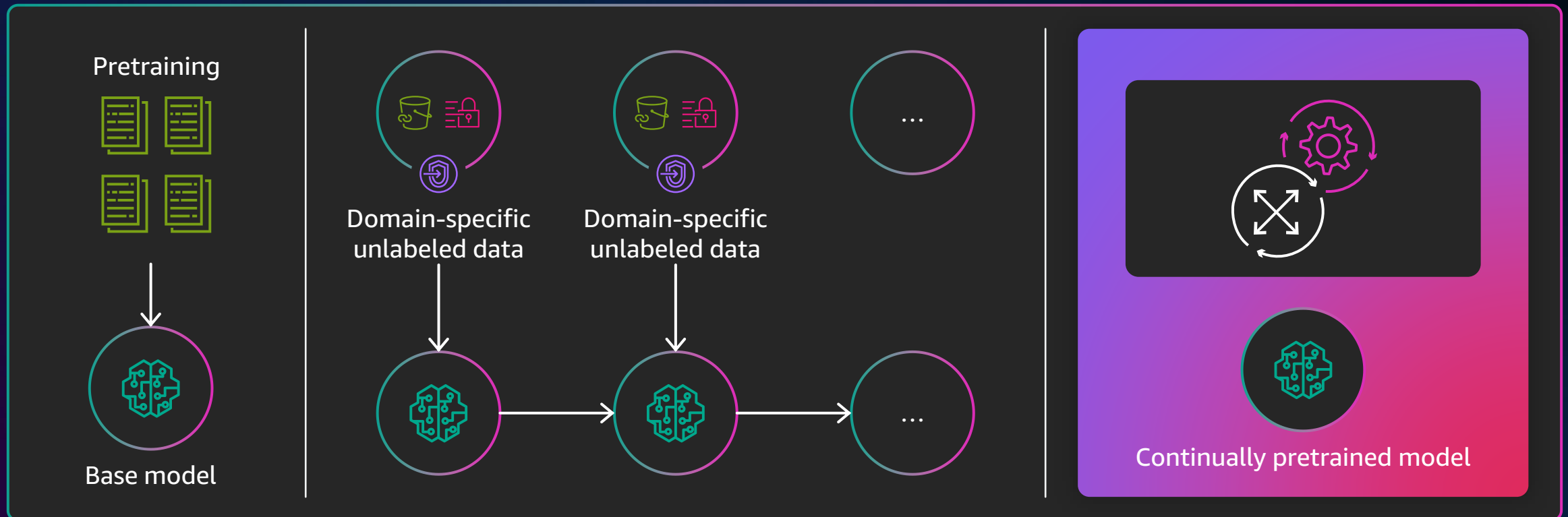
PREVIEW



Customize
models

Continued pretraining

Adapt model responses to the vocabulary
and terminology specific to your domain



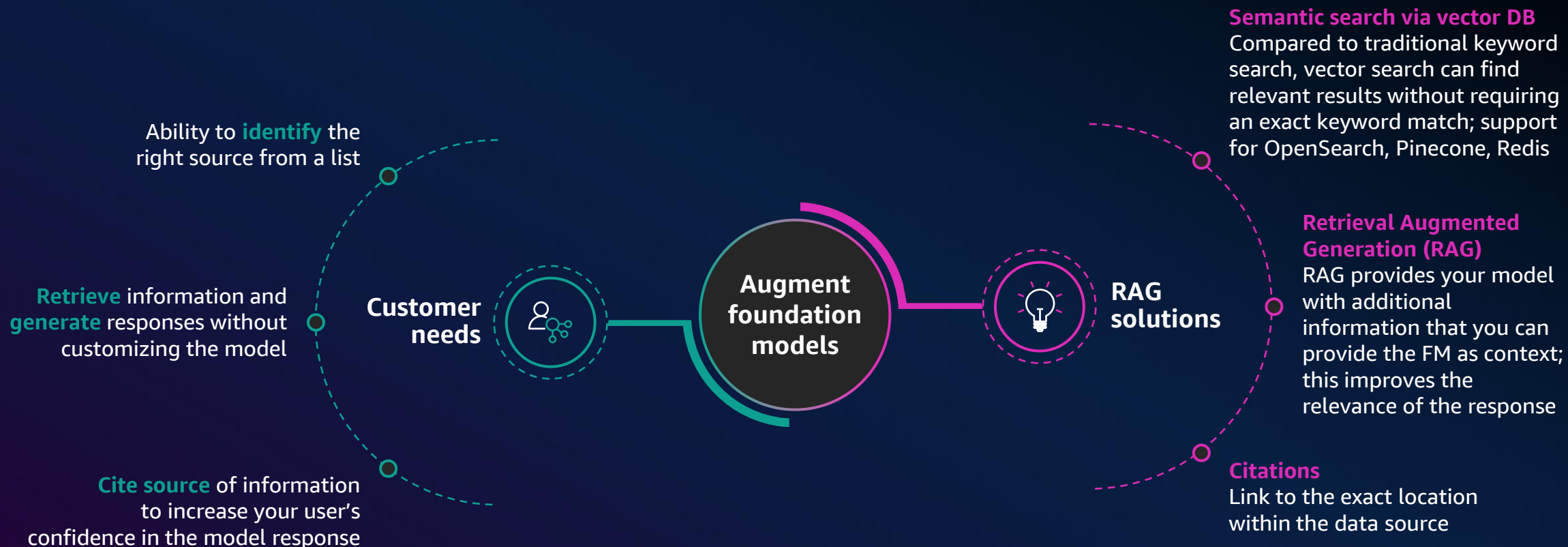


Knowledge bases/RAG

Augmenting FMs with retrieved information

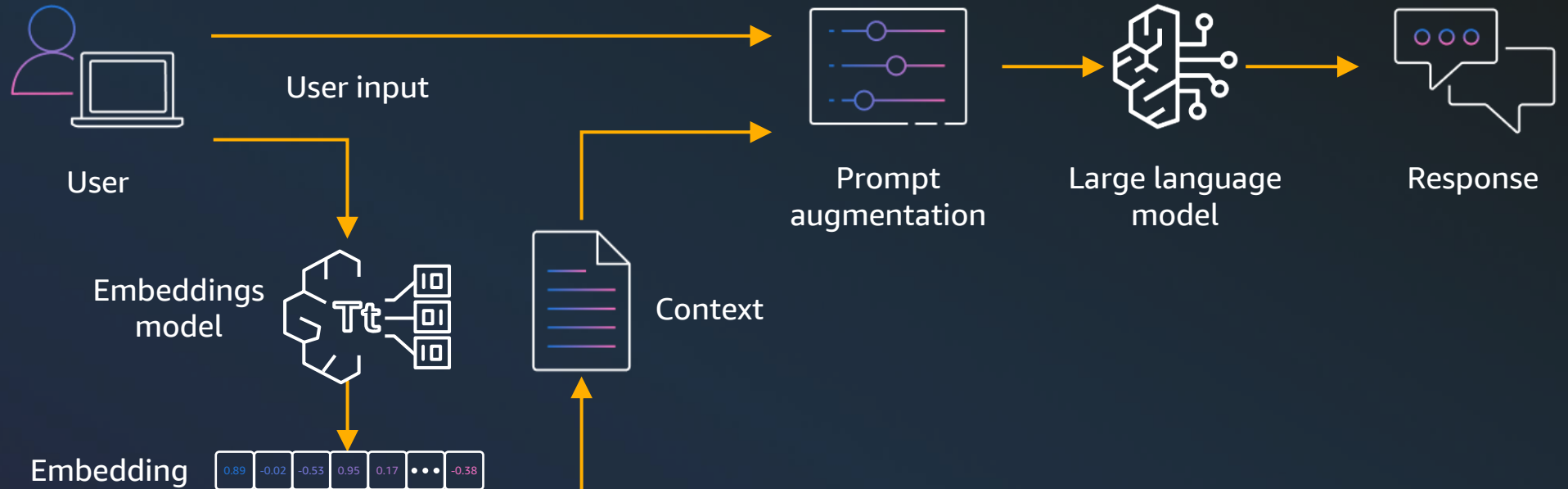


In-context
learning
and embeddings

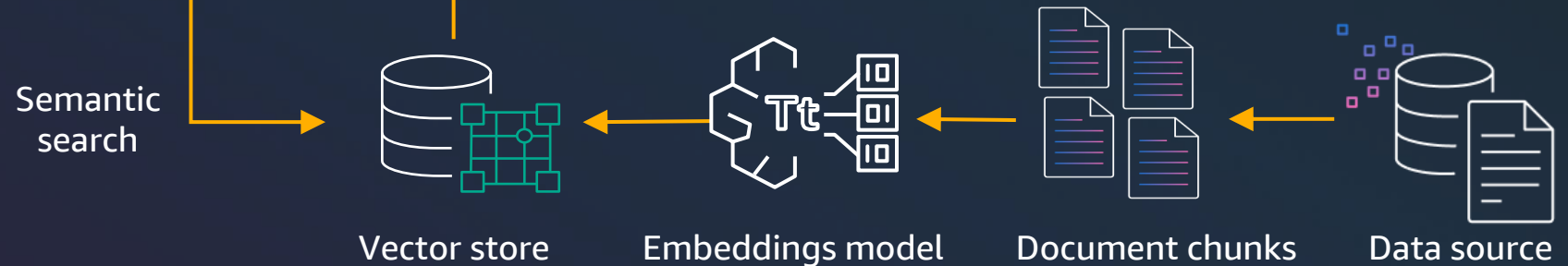


RAG in action

Text generation workflow

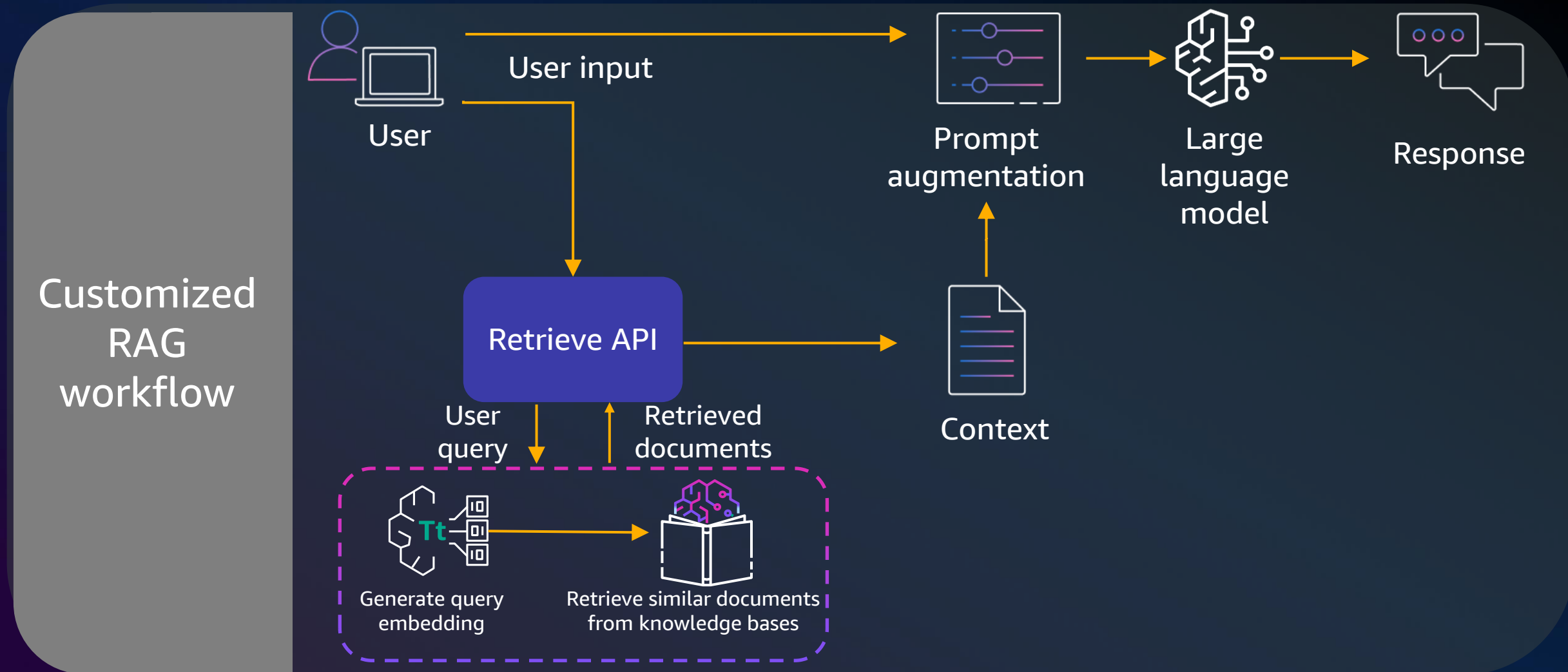


Data ingestion workflow



Customize RAG workflows using Retrieve API

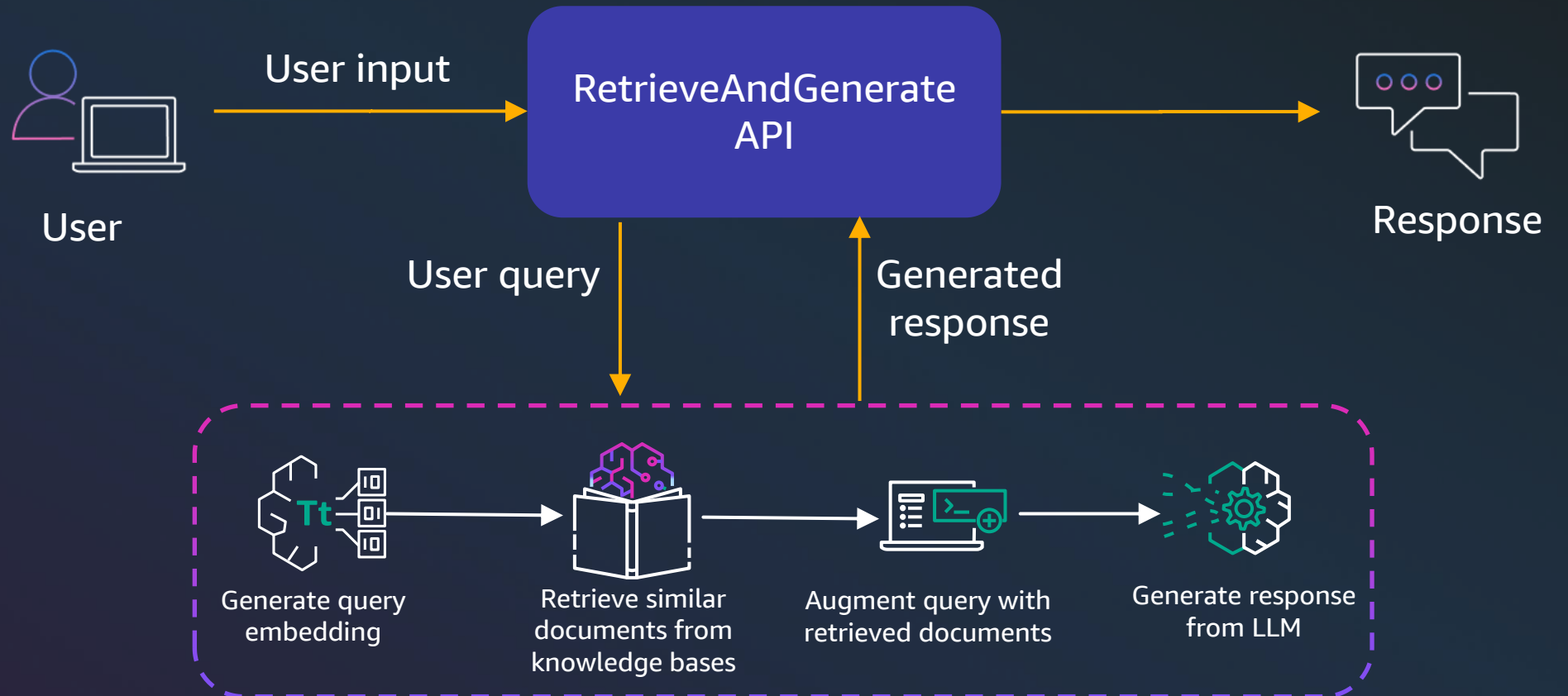
KNOWLEDGE BASES FOR AMAZON BEDROCK

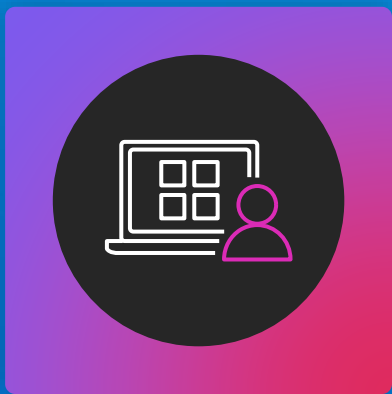


RetrieveAndGenerate API

KNOWLEDGE BASES FOR AMAZON BEDROCK

Fully
managed
RAG





Agents

Extending FMs to perform tasks



Extending FMs to performs tasks requires complex prompt engineering and cumbersome integration



Developers need to invest in prompt engineering to enable orchestration and enable access to external systems via APIs



Apps have to respond to user queries with up-to-date information by looking up data from the company's data sources



Developers have to provision and manage the infrastructure required for interaction between systems, data, and FMs

Agents on Amazon Bedrock



Prompt
engineering
agents

Accelerate delivery
of generative
AI applications
with automated
orchestration and
easy integration



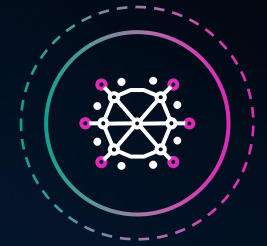
AUTOMATED PROMPT ENGINEERING

Automatically builds
the prompts based on
the information and
actions provided to enable
orchestration of tasks



SECURE ACCESS TO COMPANY SYSTEMS AND DATA SOURCES

Securely augments it
with your company-specific
information to provide
responses back to the
user in natural language



FULLY MANAGED INFRASTRUCTURE

Seamless support
for monitoring, encryption,
user permissions, and API
invocation management

How it works: Agents



Prompt
engineering
agents



Prompt editor agents



▼ Advanced prompts

Advanced prompts are a set of system-generated combination of the instruction to the model, action groups, and knowledge bases. You can edit them to override and enhance existing configurations.
[Learn more about formatting advanced prompts](#)

Select parser Lambda function - *optional*
Lambda will parse and update the LLM output so that the API can use it as input.
Select a previously created Lambda function or visit [AWS Lambda](#) to create a new function.

Select a Lambda function

\$Latest

↻

Start

Orchestrate

Retrieve

Stop - disabled

Orchestration prompt info

This prompt defines the order in which actions are executed.

Human: You are a helpful assistant for general questions about amazon benefits
Given a set of APIs, Conversation History, Attributes and User Input, create an orchestration plan for executing the set of apis in order to fulfill user input.
Valid "api" values are
,x_amz_knowledgebase_T4NGYQKBJJ::Search or "Cannot Answer" or "Final Answer"
DO NOT return an api if all required parameter values are not present.
DO NOT replace the placeholders in the api_name with api_inputs.
Valid "verb" is HTTP verb used in "APIs" e.g. GET, PUT etc
Valid "api_input" as json from "User Input", "Observation" or "Conversation History" or "Attributes".
NEVER assume value for any parameter, mark the value as "null" if not available.
DO NOT ask information which is already collected previously and is available in "Conversation History"
DO NOT ask information if provided in the "Attributes".
- For each parameter in "api_input" extract word-for-word quotes from "User Input", "Conversation History", "Attributes" or "Observation" in "source" from where the value was mapped. If there is no relevant quote to extract, mark value and source as "null".
- ALWAYS convert the "api_input" to format mentioned for the parameter in the API . eg. if a parameter is number type, convert it to number representation.
DO NOT go into a loop and return exact same apis with exact same api_input as previous observation
Provide only ONE action per \$JSON_BLOB, as shown below and always stick to the below format:
...
{
 "api": \$API_NAME,
 "verb": \$HTTP_VERB,
 "api input": { \$PARAMETER: {"value": \$INPUT, "source":

✎ Edit the orchestration prompt

Override this prompt's default configurations

Orchestration prompt configurations

☒ Enable this prompt for predictions

☐ Use Lambda function for parsing

Inference configurations

▼ Randomness & Diversity info

Temperature

1

Top P

1

▼ Length info

Max completion length

100

Stop Sequences

Add

A stylized icon representing the ReAct framework, showing a person's head with a speech bubble and a star, indicating reasoning and action.

ReAct

FMs can reason and figure out how to solve user-requested tasks with a reasoning technique called ReAct

A stylized icon representing the Chain of Thought framework, showing a flowchart with a diamond shape, indicating a sequence of reasoning steps.

Chain of thought

The thought is a reasoning step that helps demonstrate to the FM how to tackle the problem and identify an action to take. The action is an API that the model can invoke from an allowed set of APIs. The observation is the result of carrying out the action.

The AWS logo, consisting of the letters 'aws' in a stylized font with a curved line underneath.

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Chain of thought (CoT) trace



Amazon Bedrock

✕

Test

AgentTestAlias: DRAFT ▾

Getting started

Overview

Examples

Foundation models

Base models

Custom models

Providers

Playgrounds

Chat

Text

Image

Orchestration

Knowledge base [Preview](#)

Agents [Preview](#)

Deployment

Provisioned throughput

Model access

Settings

User guide [🔗](#)

Bedrock Service Terms [🔗](#)

Can you help send a reminder to all pending claims

To proceed with claim-123, please provide the Accident Report, Driver's License and Vehicle Registration. A reminder has been sent for the pending documents.

< Hide trace

Trace (5)

Request ID: a45685fa-c208-45ac-b929-8b54fd0f352c [📄](#)

▼ Step 1

Success

Trace

1 {

2 "rationaleTrace": {

3 "text": "Based on \"User Input\" I need to send reminders for

4 all open claims. I will first get list of all open claims

5 using ClaimManagementActionGroup:getAllOpenClaims API",

6 "traceId": "7ccacf6f-5e35-423b-a29d-57b612b783eb-0"

7 },

8 "invocationInputTrace": {

9 "actionGroupInvocationInput": {

10 "actionGroupName": "ClaimManagementActionGroup",

11 "apiPath": "/claims",

12 "verb": "GET"

13 },

14 "invocationType": "ACTION_GROUP",

15 "traceId": "7ccacf6f-5e35-423b-a29d-57b612b783eb-0"

16 },

17 "observationTrace": {

18 "actionGroupInvocationOutput": {

19 "text": "{ 'response': [{ 'claimId': 'claim-123',

20 'policyHolderId': 'A134085', 'claimStatus': 'Open'},

21 { 'claimId': 'claim-06', 'policyHolderId': 'A645987',

22 'claimStatus': 'Open'}] }"

23 },

24 "invocationType": "ACTION_GROUP",

25 "traceId": "7ccacf6f-5e35-423b-a29d-57b612b783eb-0"

26 }

Thought:

I need a list of claims

Actions:

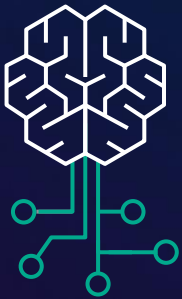
API Claims GET

Observation:

List of claims



Amazon Bedrock simplifies



Choice



Customization



Integration

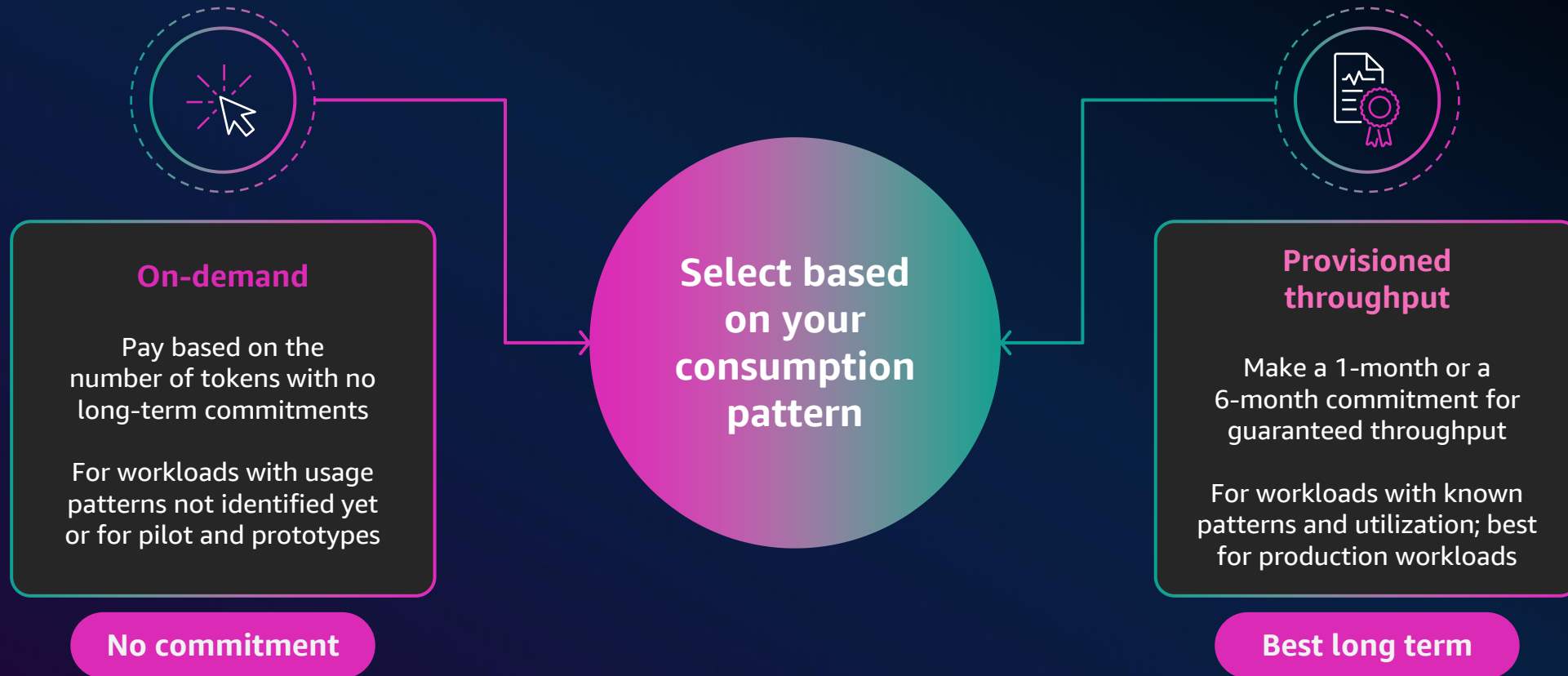


Responsible AI

Consumption choice



Integrate into
your stack



Provisioned throughput



Integrate into
your stack

Allocate model units

Engage with AWS to allocate model units for a specific model

Each model unit provides a predetermined level of guaranteed throughput

Purchase throughput

Once allocated, purchase throughput directly from the console

Guaranteed throughput for the term of commitment—option to renew automatically

Amazon Bedrock > Provisioned throughput > Purchase provisioned throughput

Purchase provisioned throughput [Info](#)

Provisioned throughput details [Info](#)

Provisioned throughput name

Name can have up to 40 characters, and it must be unique. Valid characters A-Z, a-z, 0-9, and - (hyphen).

Select model

Select category

Select model

--

► Tags - optional

Model units & commitment term [Info](#)

Select model units & commitment term to purchase Provisioned throughput. To estimate cost use [MU Estimator](#).

Model units

Please request the model units here before purchasing provisioned throughput. [AWS support center](#)

Select commitment term

Commitment terms locks the purchase for the selected duration.

Select commitment term

Estimated purchase summary

To view the provisioned throughput pricing please visit [Pricing information](#)

Estimated hourly cost	Estimated daily cost	Estimated monthly cost
—	—	—

Edits to model and model units will be restricted

Once provisioned throughput is purchased, model units cannot be updated and the model can only be updated to another model with the same lineage.

[Learn more](#)

Cancel

Purchase Provisioned throughput

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Compute optimization



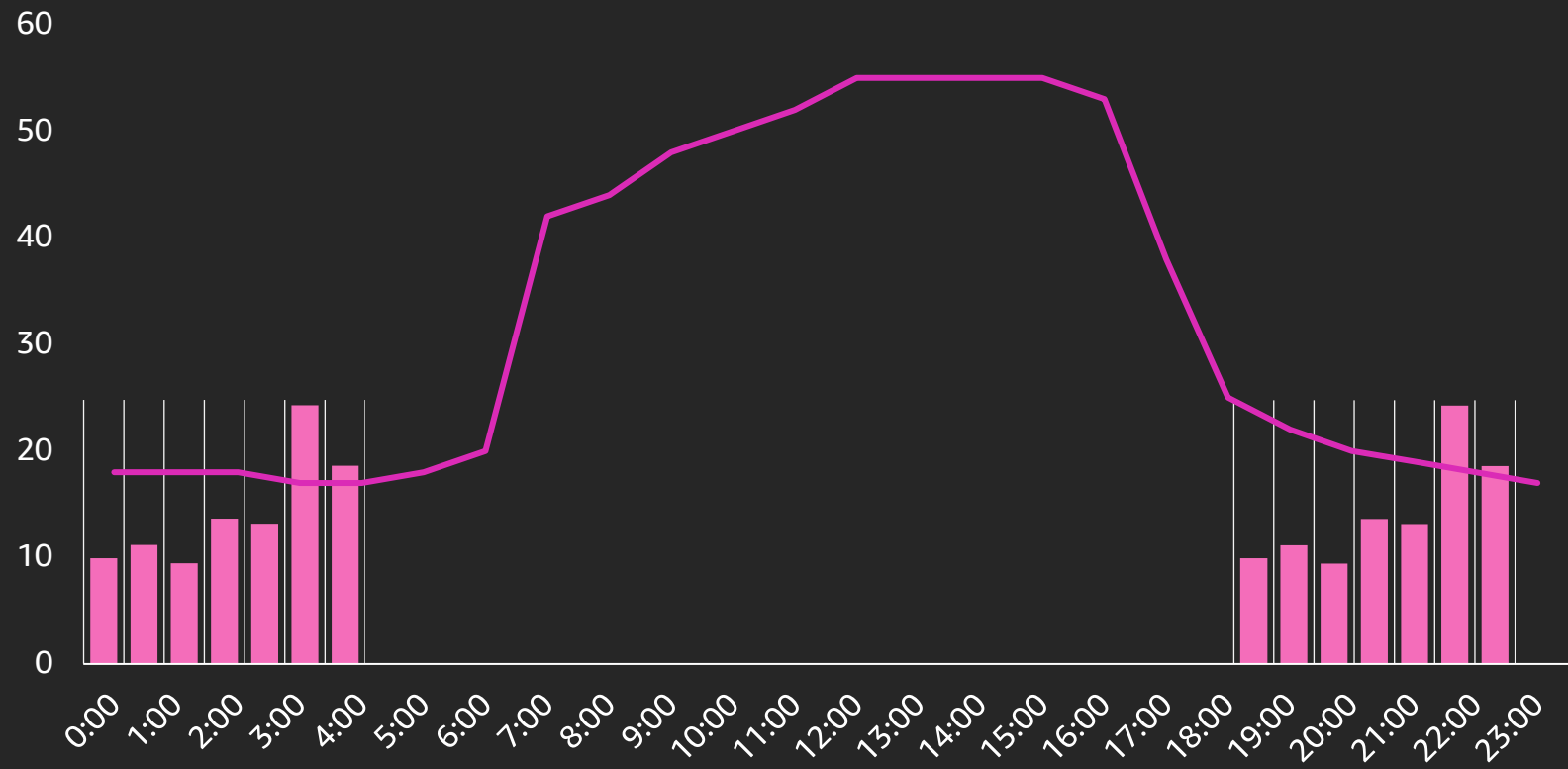
Integrate into
your stack

IN PREVIEW

Batch Async API

Easily execute batch
workloads during
non-peak hours

Average requests per hour

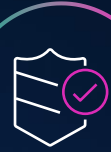




Guardrails

Guardrails for Amazon Bedrock

Implement application-specific safeguards based on your use cases and responsible AI policies



Apply guardrails to multiple foundation models and agents for Amazon Bedrock



Configure harmful content filtering based on your responsible AI policies

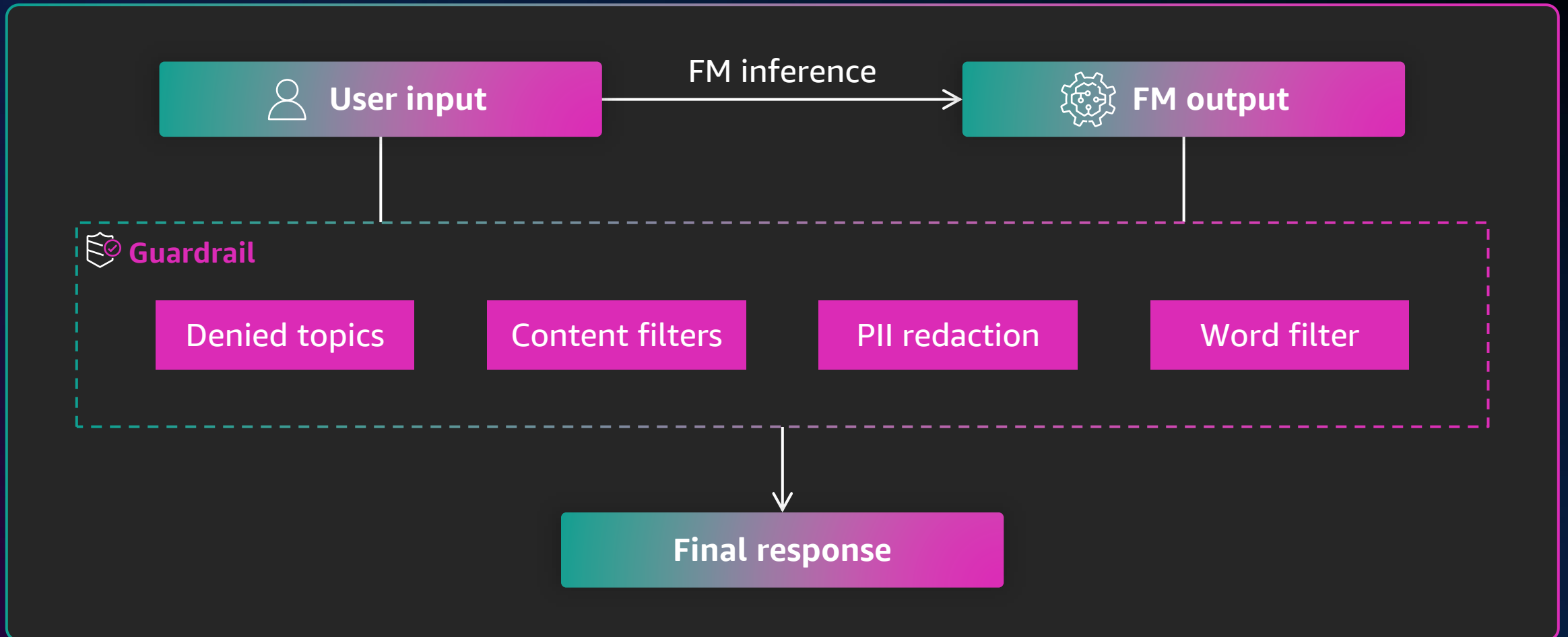
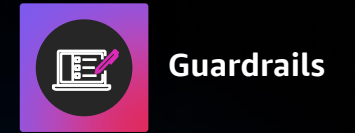


Define and disallow denied topics with short natural language descriptions



Redact sensitive PII in FM responses

How it works



Guardrails



Guardrails

1

Amazon Bedrock > Guardrails > Create guardrail

Step 1
Provide guardrail details

Step 2 – optional
Add denied topics

Step 3 – optional
Configure content filters

Step 4
Define blocked messaging

Step 5
Review and create

Add denied topics

Denied topics are topics that you don't want your end users to have with the FM or agent.

▼ Denied topic 1

Clear

Delete

Name

Enter name

Definition for topic

Outline how model should use this topic.

Enter instructions

Valid characters are a-z, A-Z, 0-9, _ (underscore) and - (hyphen). This instruction can have up to 150 characters.

Sample utterances - optional

Representative phrases that you expect a user to type to invoke this topic. Please enter 3-5 utterances.

Add new utterance

Add topic

Cancel

Previous

Next

2

Amazon Bedrock > Guardrails > Create guardrail

Step 1
Provide guardrail details

Step 2 – optional
Add denied topics

Step 3 – optional
Configure content filters

Step 4
Define blocked messaging

Step 5
Review and create

Configure content filters

Define filtering thresholds that are right for your use case. By using these toxicity filters, you can take comprehensive measures to block content that violates your usage policies.

Filter strengths for prompts

A higher filter strength will filter more content.

Reset

Enable filters for prompts

Hate

Description

NoneLowMediumHigh

Insults

Description

NoneLowMediumHigh

Sexual

Description

NoneLowMediumHigh

Violence

Description

NoneLowMediumHigh

Filter strengths for responses

A higher filter strength will filter more content. These filters filter out content from model response; they do not modify model behavior.

Reset

Enable filters for responses

Hate

Description

NoneLowMediumHigh

Insults

Description

NoneLowMediumHigh

Sexual

Description

NoneLowMediumHigh

Violence

Description

NoneLowMediumHigh

Model responses classified with HIGH or MEDIUM confidence will be filtered.
Model responses classified with NONE or LOW confidence will be allowed.

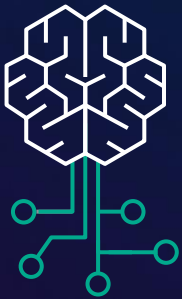
Cancel

Previous

Next

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Amazon Bedrock simplifies



Choice



Customization



Integration



Responsible AI

Data privacy & security



Securely build generative AI applications with your data



Build with comprehensive data protection and privacy;
use AWS security services and best practices

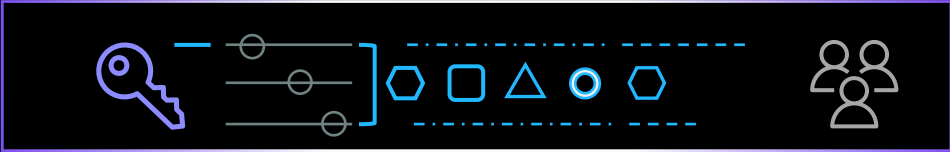
Data privacy & security



Data privacy
& security



AWS Identity and Access
Management (IAM)



Customer managed keys



PrivateLink